

Airport Management System

Build Info

The folder includes the GUI and Backend files, as well as the airport.db provided. Here's how to build and run.

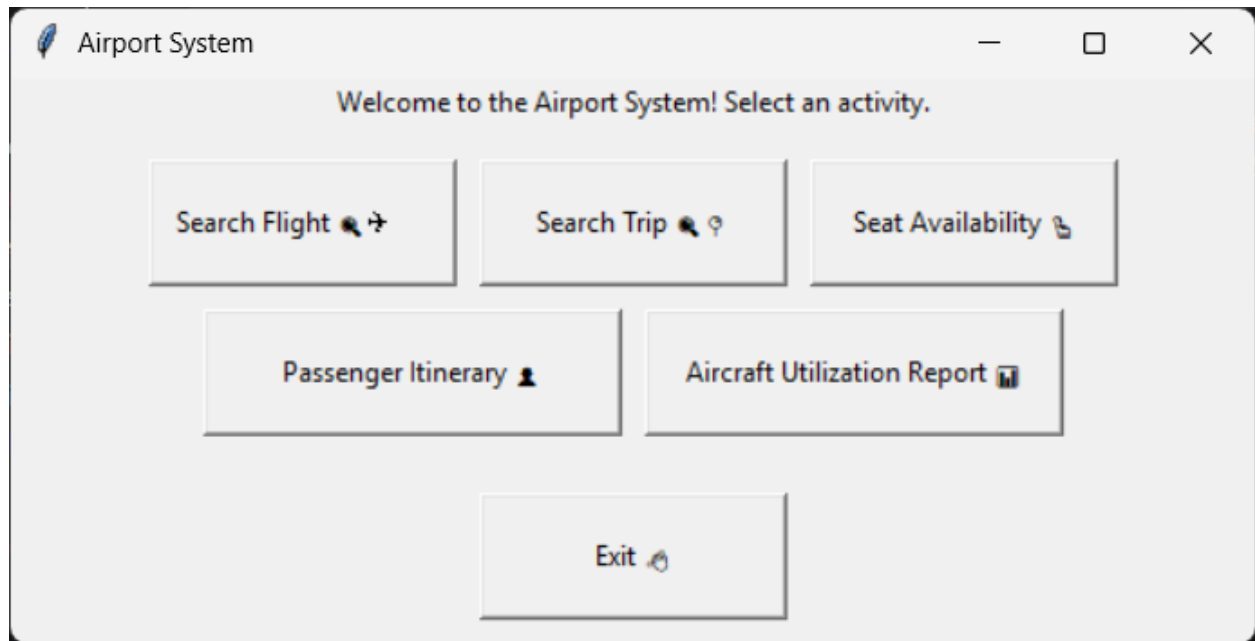
1. Open an IDE or any code editor (recommended: Visual Studio Code)
2. Open an entire folder into the IDE/code editor, making sure the program is fully aware of the scope of the project.
3. Go to the GUI folder, select Main_Menu.py, and simply run it from there (running any other GUI code snippet will jump straight into the program's features. Clicking back would bring back to the Main Menu.)
4. Have a blast! When you're done, click "Exit".

Design Patterns

Part of the attached file for submission is a picture of what the GUI would typically look like planned out on Microsoft Whiteboard. Mentioned earlier, the program is organized into Backend files (SQL files), and GUI files (Python files with tkinter package). For the GUI framework, we used tkinter package; back in Milestone 2, the project was coded in Python, and we discovered that we could easily build a GUI on top of this language using this package. It comes built in with the language, no extra installs. It was easy to setup. The reason the GUI is designed the way it is is that one of the teammates, when taking class notes, usually group their information into literally overlapping information blocks; that method was essentially carried over to the design of the GUI, and we branched off from there. Since most of the coding was based on the logic used in the last Milestone, a lot of the GUI menus are essentially big white output boxes that'll output the results for you.

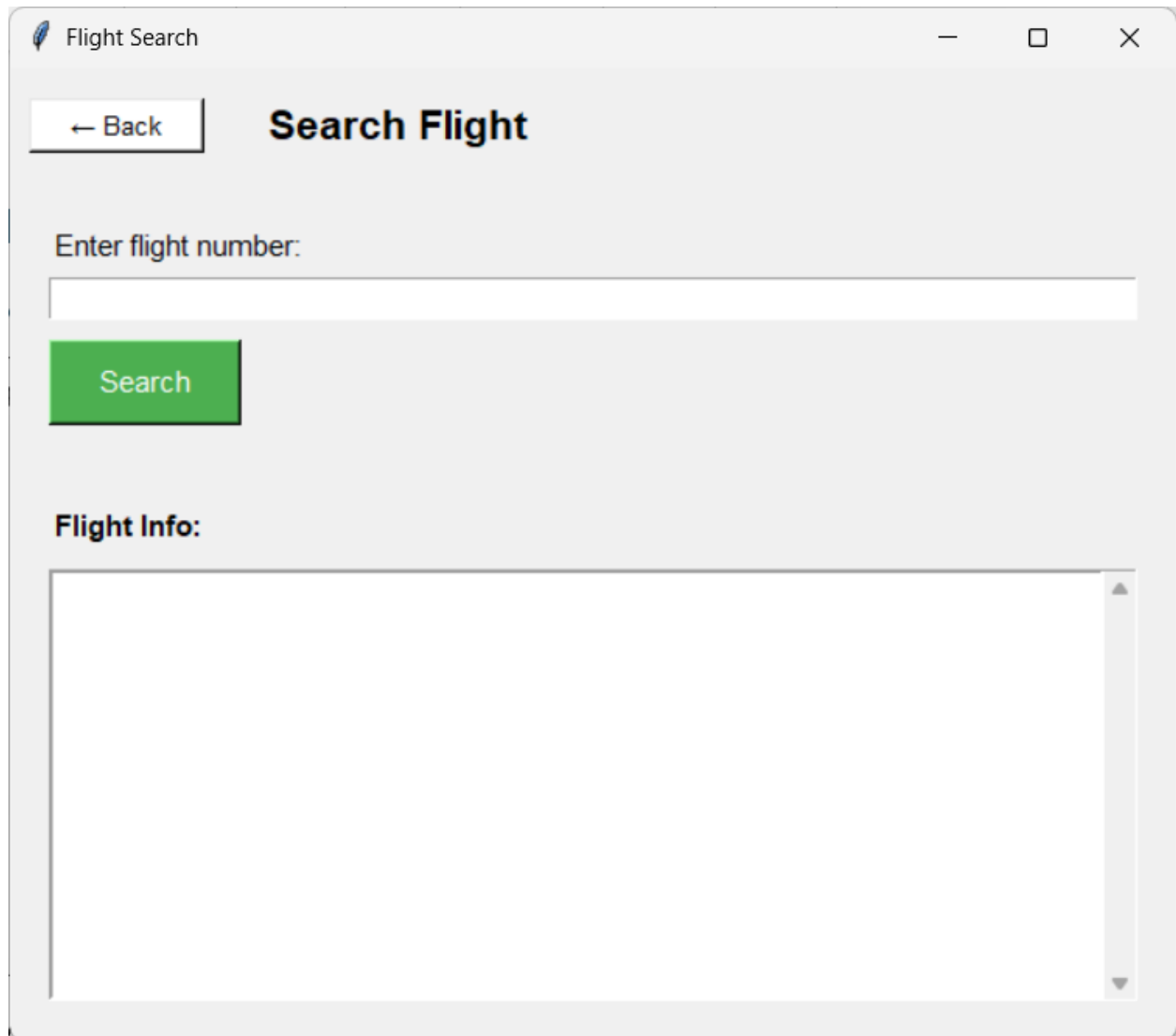
Quick Start Guide

Main Menu



Pick any option you want to do!

Search Flight



The image shows a window titled "Flight Search" with standard window controls (minimize, maximize, close) in the top right corner. Inside the window, there is a header area with a "← Back" button on the left and the title "Search Flight" in bold. Below the header, there is a label "Enter flight number:" followed by a text input field. Underneath the input field is a green button with the text "Search". Below the search section, there is a label "Flight Info:" followed by a large, empty rectangular area with a vertical scrollbar on the right side, intended for displaying search results.

Enter any flight number (format: [letter][letter][four numbers]), and the results will show the information.

Search Trip

 Search Trip

← Back

Search Trip

From airport:

To airport:

Search Trip

Flight Info

Choose any two airports (abbreviated by three letters), and you'll be shown the possible paths you can take, whether single or multiple legs.

Seat Availability

Seat Availability

← Back

Seat Availability

Flight number:

Date (YYYY-MM-DD):

Check

Available Seats

Leg no: Seat no: Customer name: Phone:

Book This Seat

Pick a flight number and a date, and the results will show any available seats.

With the list shown, you can pick a leg and seat number, input your Name and Phone Number, and you can book that seat!

Passenger Itinerary

 Passenger Itinerary

[← Back](#)

Passenger Itinerary

Search passenger's name:

Passenger Flights:

Search up a passenger name, and check out what flights they're in!

Airport Utilization Report

Aircraft Utilization Report

← Back

Aircraft Utilization Report

From (YYYY-MM-DD): To (YYYY-MM-DD):

Report

Select a date range, and get a list of all the flights information, from airports, seats, depart and arrival time, etc.